

# THE GAP GAME

**SECURITY AND INCETIVES IN  
PROOF-OF-WORK**

A CRITICAL ANALYSIS OF THE EVOLUTION FROM  
BLOCK SUBSIDY TO TRANSACTION FEES

MICHAEL CIMARELLI 2219492

TOMMASO RINALDI 1983126

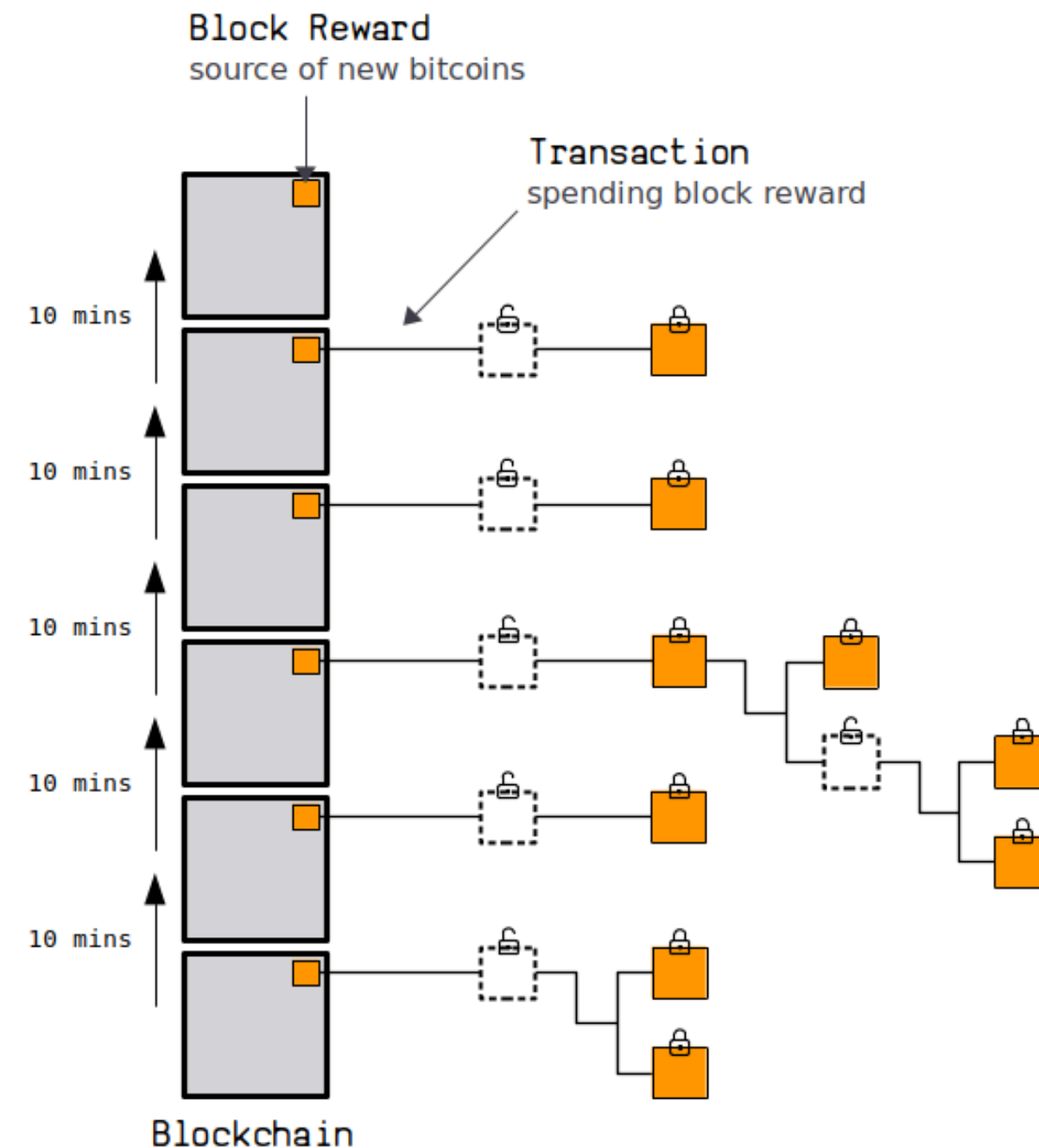


**SAPIENZA**  
UNIVERSITÀ DI ROMA

# Blockchain Incentives

PoW miners compete to solve cryptographic puzzles to validate blocks. Their reward consists of two components:

- **Block Subsidy:** Newly minted coins (currently dominant in networks like Bitcoin).
- **Transaction Fees:** Paid by users. These are destined to become the primary incentive as subsidies halve over time.

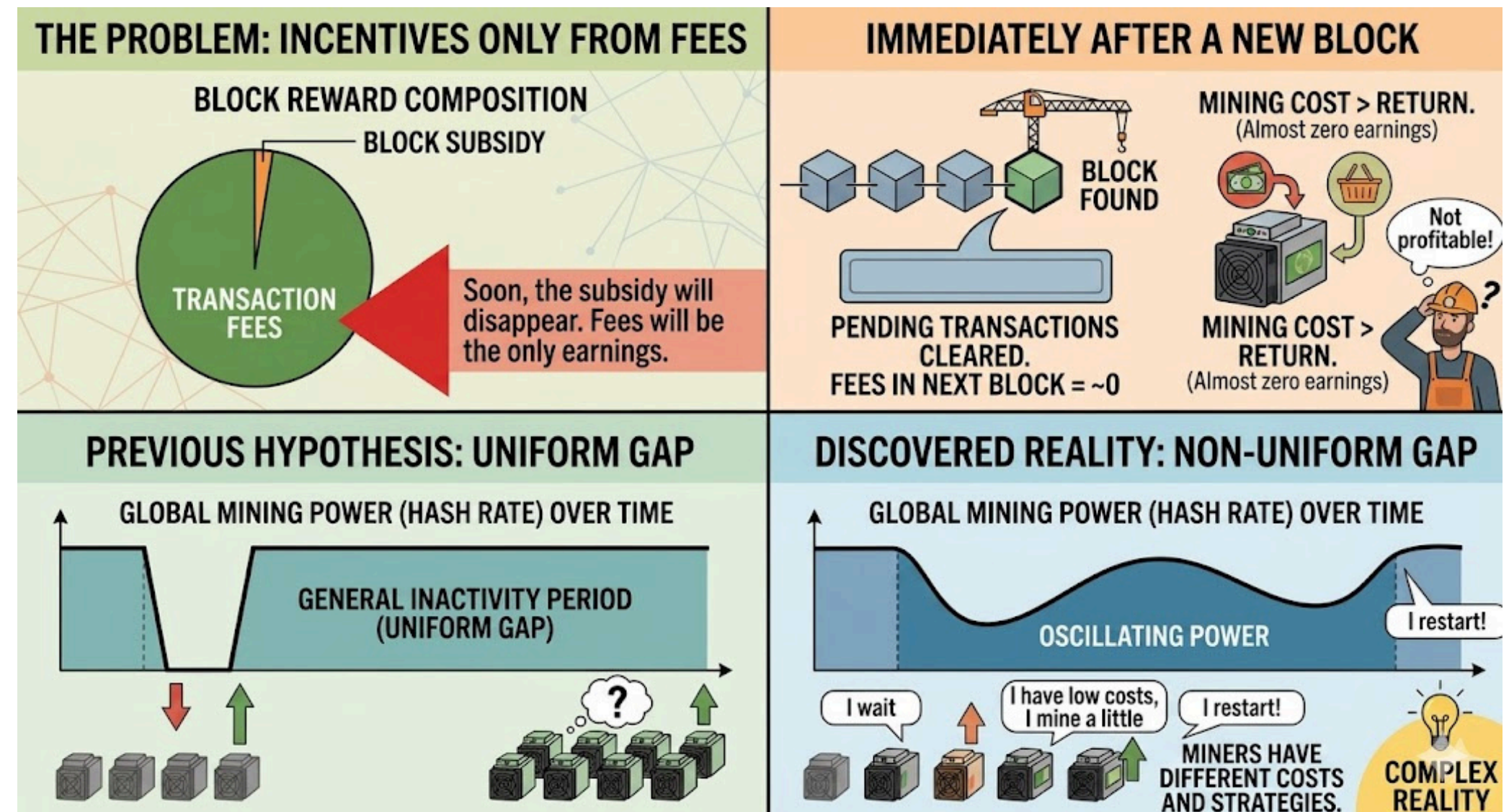


# The "Mining Gap" Problem

## Rational Inactivity

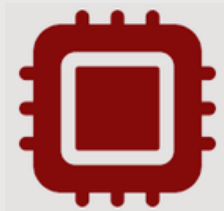
Immediately after a block is found, the mempool is cleared. The available fees drop to near zero.

For a rational miner, the Operational Cost (Opex) exceeds the potential reward at that moment.



**Result:** Miners turn off machines and wait for fees to accumulate. This period of inactivity is the "Mining Gap".

# | Modeling the Gap Game



## CAPEX

Capital Expenses: Fixed costs of hardware ownership. These are paid regardless of whether the machine is on.



## OPEX

Operational Expenses: Electricity and cooling. These are only paid when the miner is active.



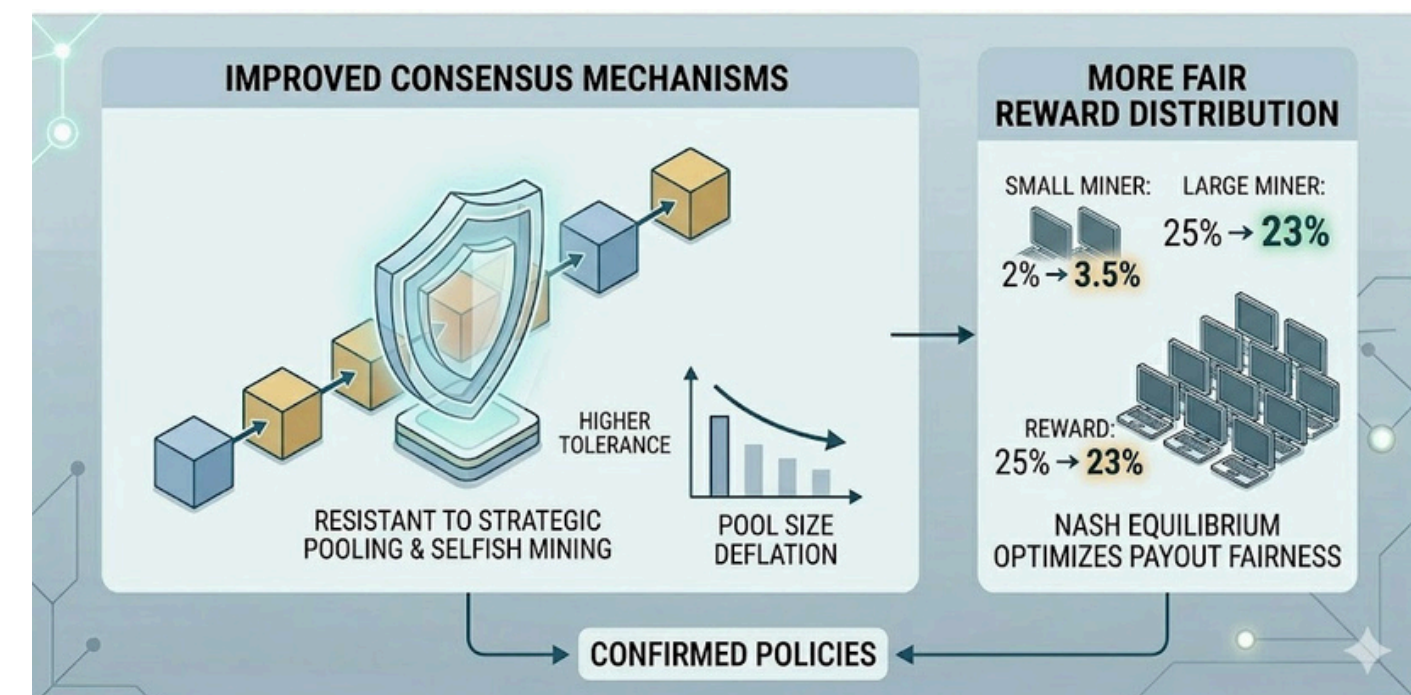
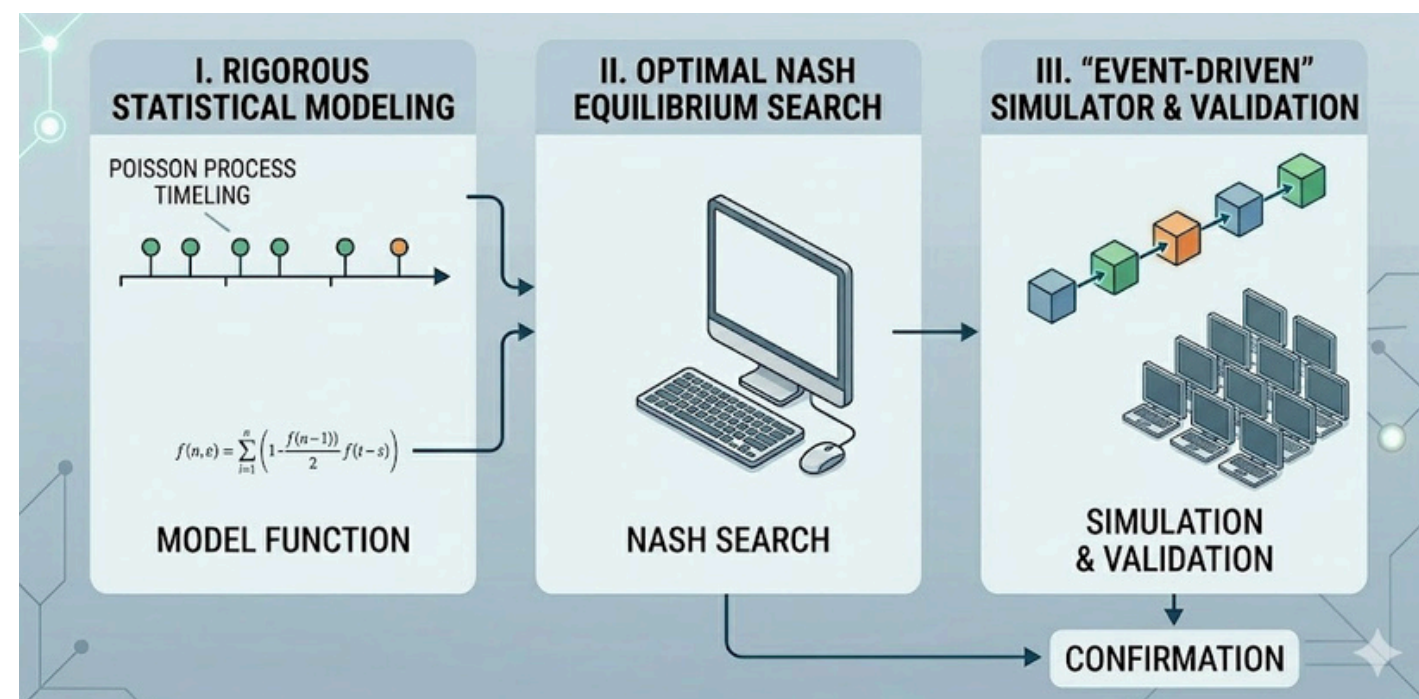
## STRATEGY

Miners decide the exact "Start Time" to maximize profit, balancing the risk of losing a block vs. saving on Opex.

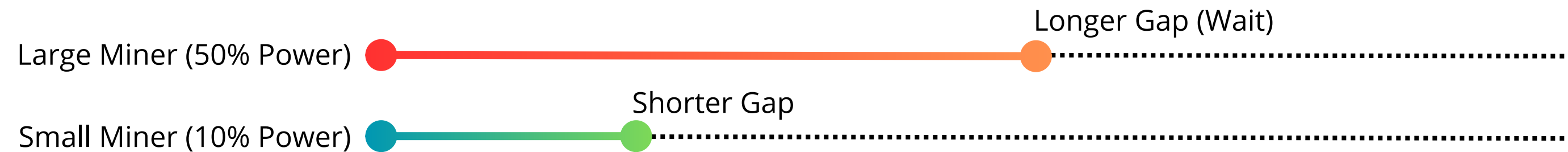


# Research Methodology

|   | Phase                | Approach                             | Key metric/Goal                  |
|---|----------------------|--------------------------------------|----------------------------------|
| → | Statistical Modeling | Memoryless Exponential Distributions | Calculate Utility Functions      |
| → | Numerical Search     | Numerical Analysis Tools             | Find $\epsilon$ -Nash Equilibria |
| → | Validation           | Event-driven Simulator               | Test against thousands of blocks |

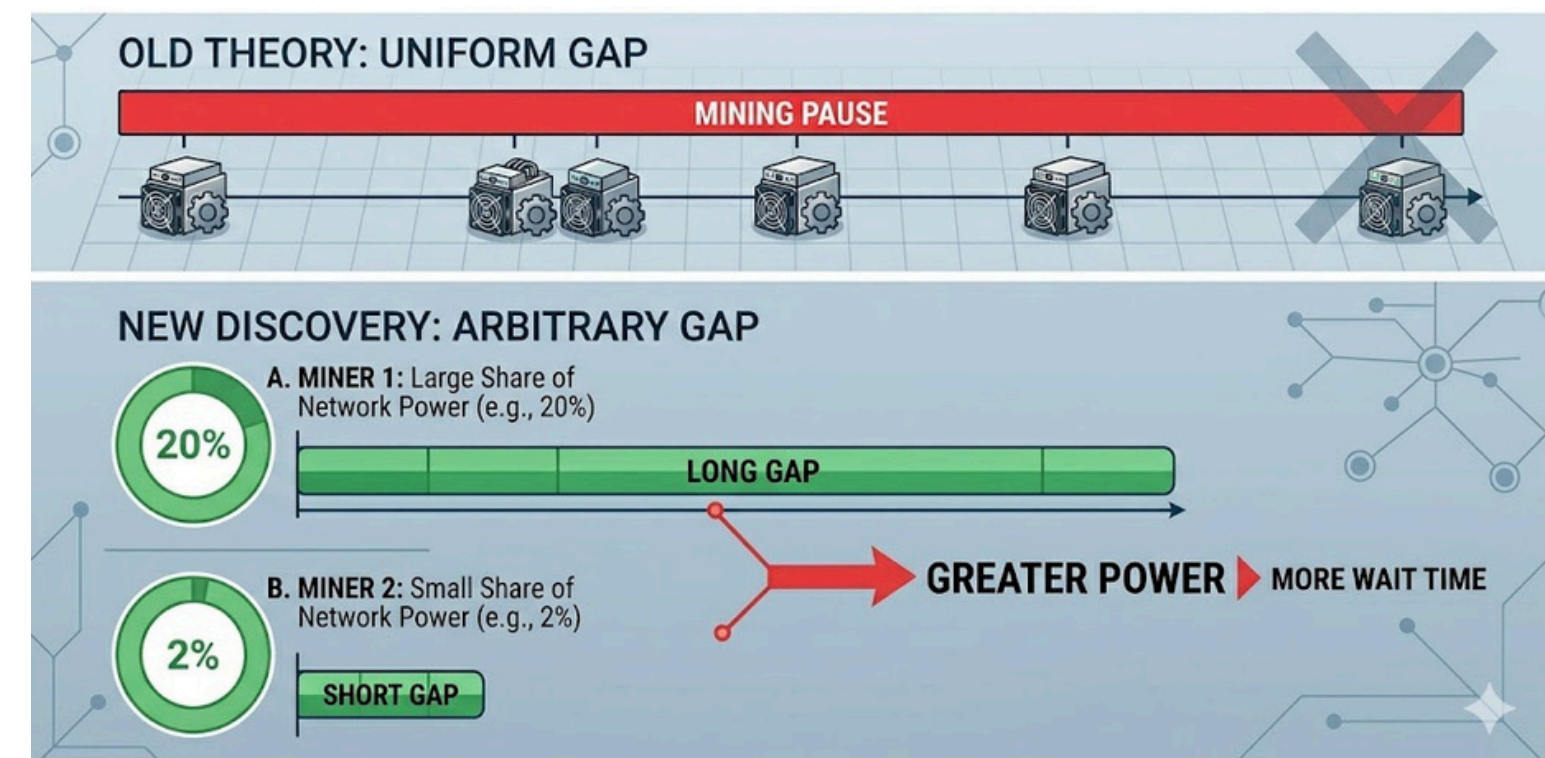


# The Asymmetric Gap Result



Contrary to previous theories of a "uniform gap," the study discovers **Arbitrary Gaps**.

Large miners delay activation longer because they can sacrifice a small win probability to achieve massive savings on Opex.





# Centralization Pressures

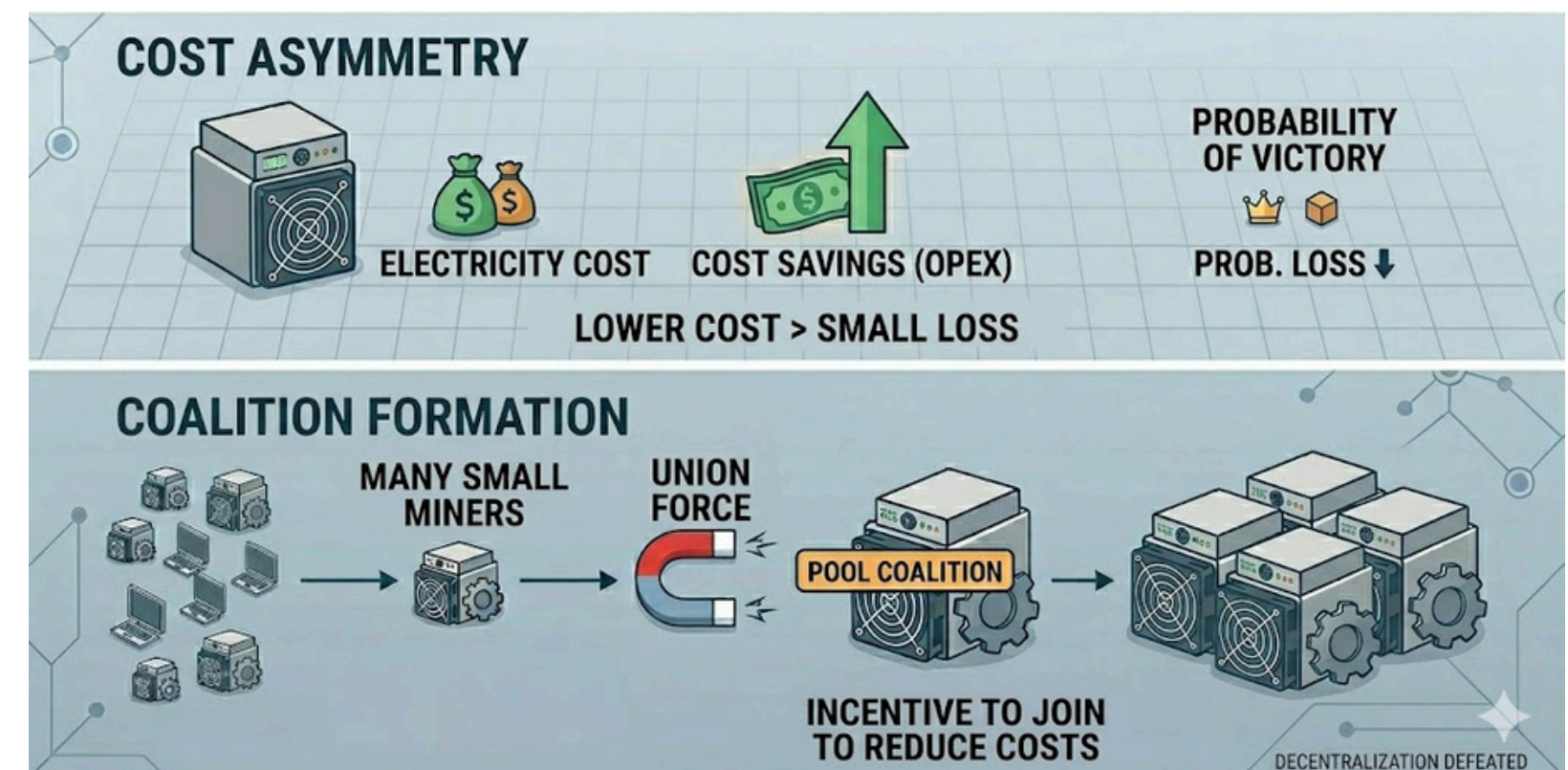
# 90%

Power Drop Possible

## The Scale Advantage

Because the "Gap Game" favors those who scale their operations, small miners are mathematically incentivized to merge.

This destroys the foundational principle of blockchain:  
**Decentralization.**



# 51% Attack Vulnerability

## ⚠ Network Under Threat

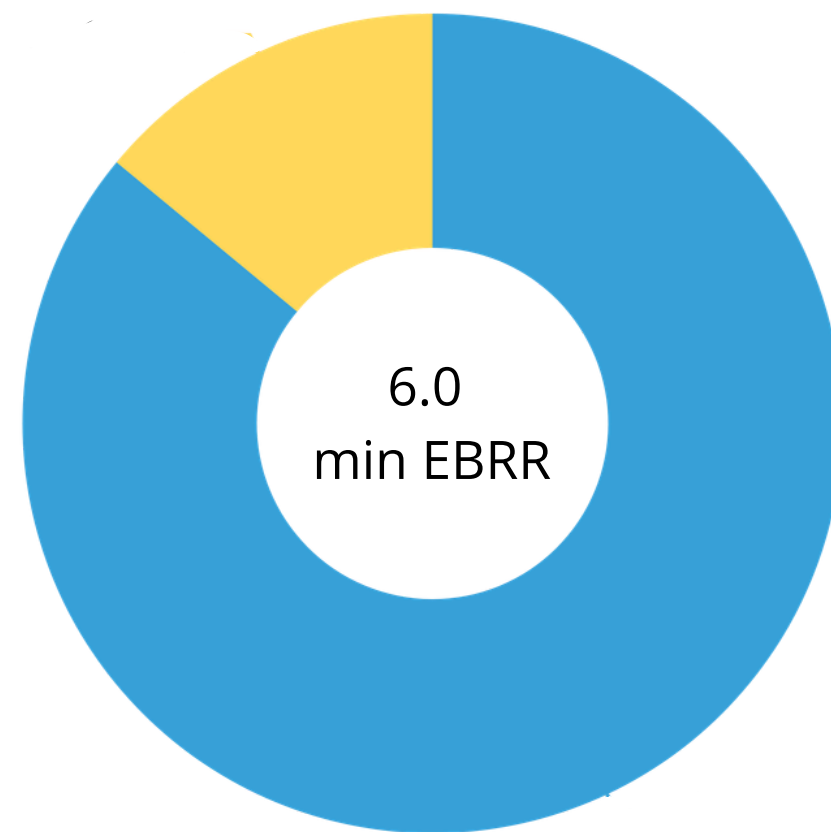
During a mining gap, the effective hash power protecting the network crashes. In extreme scenarios (low subsidy + high Opex), power utilization fell to **10%**.

An attacker would need far fewer resources to launch a **51% Attack** and manipulate the ledger during these vulnerable windows.





# Proposed Solution: EBRR



## Expected Base Reward Ratio

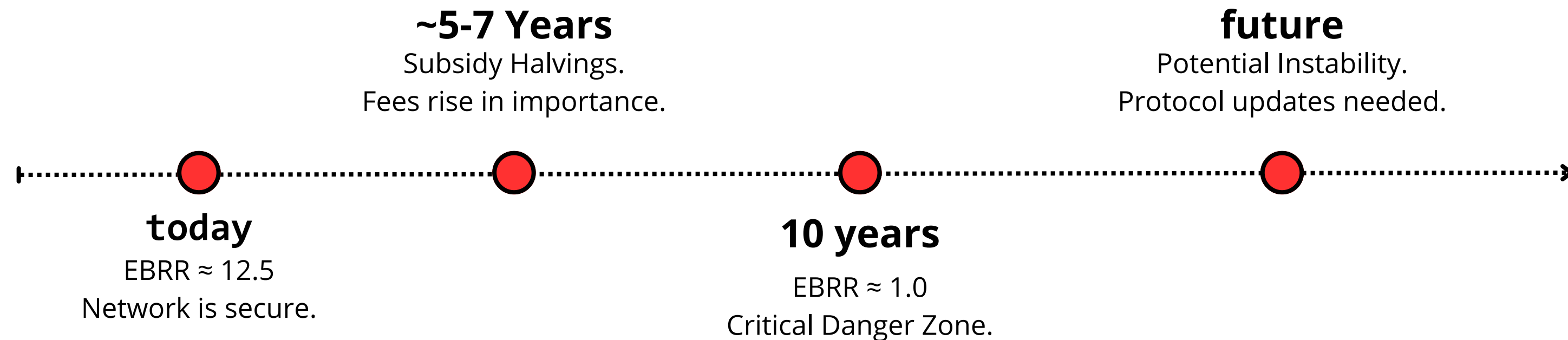
To avoid gaps, the network must guarantee a base reward that justifies Opex immediately after a block.

-   $EBRR \geq 6$  (Secure Zone)
-   $EBRR < 1$  (Critical Danger)

$$EBRR = \frac{\text{Block Subsidy} + \mathbb{E}[\text{Minimum Fees}]}{\text{Average Operational Cost (Opex)}}$$



# Bitcoin Timeline & Conclusions



## Final Verdict

Blockchain protocols must integrate Fee Management Solutions now to preserve security and decentralization for the next decade.

- Fee Smoothing Protocols: Temporal redistribution of transaction fees over N subsequent blocks to linearize mining revenues.
- Adaptive Difficulty DAA: Dynamic anchoring of the cryptographic difficulty to the economic density of the mempool.